

Dealer-Gamma Feedback and Local Volatility Cycles: A Fixed-Equilibrium Bifurcation Model

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Abstract

We give a local physical-measure model in which detrended log price, square-root variance, and a filtered price signal form a feedback loop. The reference equilibrium is fixed as coupling varies and variance drift points inward at zero. For a centered Gaussian density of signed dealer positions, the Routh–Hurwitz determinant is quadratic in coupling. An illustrative, non-calibrated configuration has $\kappa^* = 31.4933 \text{ yr}^{-1}$, $\omega^* = 47.1186 \text{ rad yr}^{-1}$, and $\ell_1 = -6.2888$; integrating the actual nonlinear book gives a nearby attracting cycle with positive variance. The result is local and deterministic. Public open interest does not identify dealer sign, so a future WRDS study is pre-specified as a registered-horizon predictive proxy analysis rather than a structural confirmation.

Keywords

dealer gamma, Hopf bifurcation, stochastic volatility, open interest, pre-registration

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1 Motivation and scope

Option delta hedging creates state-dependent demand. Long-gamma rebalancing is locally contrarian and short-gamma rebalancing is locally trend-following [1, 5, 6]. Demand-based option pricing shows that intermediary constraints affect option prices [2]; it does not derive a spot drift. We therefore treat the feedback below as a reduced-form physical-measure hypothesis.

Three safeguards distinguish the model from its earlier draft. First, a centered book functional pins one equilibrium for every coupling. Second, the memory term in variance is multiplied by variance, preserving inward boundary drift. Third, theory uses latent signed dealer positions, while the empirical protocol treats public open interest as unsigned. Open interest counts outstanding contracts and does not identify participant or side [4]; participant-tagged data are needed to estimate dealer positions [7].

2 Centered physical-measure model

Let $X_t = \log(S_t/F_t)$ for a predictable reference F_t that absorbs baseline expected return. Locally,

$$dX_t = [-\delta X_t - \frac{1}{2}(v_t - \theta_v) + \kappa g(X_t, \chi_t, v_t)]dt + \sqrt{v_t} dW_t^S, \quad (1)$$

$$dv_t = [\kappa_v(\theta_v - v_t) + \gamma v_t \chi_t]dt + \xi \sqrt{v_t} dW_t^v, \quad (2)$$

$$d\chi_t = \alpha(\beta X_t - \chi_t)dt, \quad d\langle W^S, W^v \rangle_t = \rho dt. \quad (3)$$

All dynamics are under $\mathbb{P}$. At $v = 0$, the variance drift is $\kappa_v \theta_v > 0$ and diffusion is zero. This is boundary-compatible; no global non-explosion theorem is claimed.

For fixed log moneyness, a Gaussian positive book mass integrates against Black–Scholes gamma as

$$\mathcal{G}(X, v) = \frac{e^{-q_d T - X}}{\sqrt{2\pi\tau^2(v)}} \exp[-(X - m(v))^2 / (2\tau^2(v))],$$

$$\tau^2(v) = \sigma_q^2 + vT, \quad m(v) = \mu_q - (r - q_d + v/2)T. \quad (4)$$

We use

$$g(X, \chi, v) = s \frac{\mathcal{G}(X, v) - \mathcal{G}(0, \theta_v)}{\mathcal{G}(0, \theta_v)}, \quad s \in \{-1, +1\}. \quad (5)$$

Here s is a model input for signed dealer orientation, not an OI statistic. Centering gives the fixed equilibrium $z_\star = (0, \theta_v, 0)$ for every κ .

3 Local Hopf criterion

At $z_\star$, write $G_X = g_X$, $G_v = g_v$, $G_\chi = g_\chi$. The Jacobian is

$$J = \begin{pmatrix} a & b & d \\ 0 & -\kappa_v & c \\ m & 0 & -\alpha \end{pmatrix}. \quad (6)$$

Here $a = -\delta + \kappa G_X$, $b = -1/2 + \kappa G_v$, $d = \kappa G_\chi$, $c = \gamma \theta_v$, and $m = \alpha \beta$. For $\det(\lambda I - J) = \lambda^3 + c_2 \lambda^2 + c_1 \lambda + c_0$,

$$\begin{aligned} c_2 &= \kappa_v + \alpha - a, \\ c_1 &= \kappa_v \alpha - a(\kappa_v + \alpha) - dm, \\ c_0 &= -a\kappa_v \alpha - bcm - dm\kappa_v. \end{aligned} \quad (7)$$

THEOREM 1 (FIXED-EQUILIBRIUM LOCAL HOPF). *Suppose the drift is C^3 . If at $\kappa^* > 0$, $c_2, c_1, c_0 > 0$, $H := c_1 c_2 - c_0 = 0$, $H' \neq 0$, and the first Lyapunov coefficient $\ell_1 \neq 0$, then $z_\star$ undergoes a non-degenerate local Hopf bifurcation. The pair is $\pm i\sqrt{c_1}$, the third eigenvalue is $-c_2$, and*

$$\left. \frac{d \operatorname{Re} \lambda_+}{d\kappa} \right|_{\kappa^*} = -\frac{H'(\kappa^*)}{2(c_1 + c_2^2)}. \quad (8)$$

In the convention of Kuznetsov [3], $\ell_1 < 0$ gives stable small cycles on the side where the pair is unstable.

The proof factors the cubic as $(\lambda + c_2)(\lambda^2 + c_1)$ at $H = 0$, derives (8) by implicit differentiation, and applies the classical Hopf theorem. It is local: no root does not imply global stability, and a subcritical Hopf alone does not imply hysteresis.

For the static book (5), $G_\chi = 0$. Put $A = \alpha + \kappa_v$, $M = \alpha \kappa_v$, and $L = (\gamma \theta_v)(\alpha \beta)$. Then

$$H(\kappa) = G_\chi^2 A \kappa^2 + \{G_v L - G_X A(A + 2\delta)\} \kappa + A(M + \delta A + \delta^2) - L/2. \quad (9)$$

Both roots are enumerated and every positivity and transversality condition is checked. Analytic derivatives of (4) through order three enter the standard Kuznetsov multilinear formula. The tensor from the variance equation also includes $B_{v,v\chi} = B_{v,\chi v} = \gamma$.

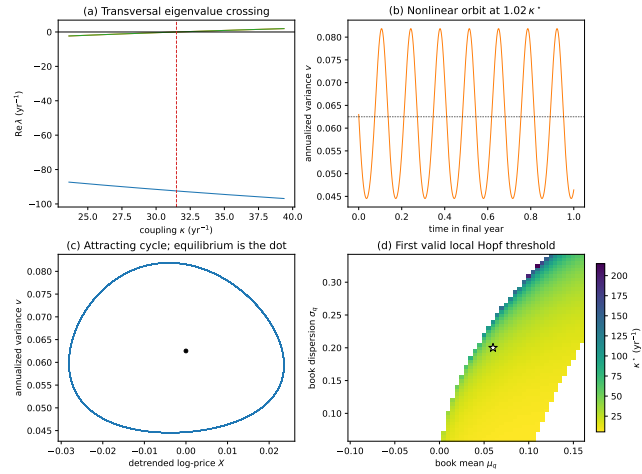


Figure 1: Eigenvalue crossing, actual nonlinear orbit at $1.02\kappa^*$, and a two-parameter local-Hopf map. Blank map cells mean no valid local Hopf root, not global stability.

4 Nonlinear validation

The illustrative parameters are

$$(\delta, \kappa_v, \theta_v, \alpha, \beta, \gamma) = (0.5, 40, 0.0625, 50, 2, 500)$$

and $(\mu_q, \sigma_q, T, s) = (0.06, 0.20, 0.25, +1)$. These are not SPX estimates. They give $G_X = -0.0617978$, $G_v = -2.0198878$ and

$$\kappa^* = 31.4932976, \quad \omega^* = 47.1185670, \quad \ell_1 = -6.2888041,$$

with crossing speed 0.2686294. The linear period is 0.13335 years. The same quadratic has a remote second valid point at $\kappa^{**} = 16860.8961$, where the pair crosses back into the stable half-plane; all figures labelled as thresholds use the first root.

Direct integration of the actual Gaussian functional, not a separately chosen cubic normal form, gives $X \in [-0.02814, 0.02358]$ and $v \in [0.04453, 0.08187]$ after transient discard. Thus the displayed cycle remains in the local and positive-variance region. The full paper's gross-normalized mixture audit is less uniform: nearby same-sign mixtures remain supercritical, but dispersed or offsetting books can be subcritical. The attracting canonical cycle is therefore not a universal book-shape prediction.

5 Pre-extraction identification protocol

No registered market dataset is analysed. Before expected September 2026 WRDS access, Amendments A13–A15 fix a registered historical horizon from 2017-01-03 through 2024-10-29. They construct four observable OI-gamma book summaries: log unsigned mass, call-put composition, gamma-weighted mean log moneyness, and log dispersion. Contracts are unadjusted standard-settlement SPX/SPXW with 1–365 calendar-day maturities, fixed quote/IV/moneyness filters, and parity forwards split by AM/PM settlement. For each, a separate forward regression predicts the next-session log squared return from the day- t summary. Leads and HAR-style lags remain on the complete CRSP market calendar; weekday indicators describe the outcome session, and one monthly-expiration control describes session t . HAC lag 5 and a

10-day moving-block bootstrap are each BH-adjusted across four tests. A label is robust only if both adjusted p -values reject and the 95% bootstrap interval excludes zero; one-family rejection is method-sensitive.

Stress-window interactions and convention-signed GEX are secondary. A convention-dependent result is labelled non-identified. Even robust prediction is association, not causation or dealer-sign identification. Participant-tagged positions and an identification strategy are required for a structural test. Because the historical outcomes and named episodes were already public, this is a retrospective pre-analysis plan, not a blinded prospective experiment.

6 Conclusion

A carefully centered, boundary-compatible three-state model can exhibit a non-degenerate local Hopf cycle. This is a possibility result, not a claim that SPX is near the threshold. The central empirical limitation is categorical: public OI is not dealer inventory. The data study is designed to stop at that boundary rather than promote a proxy correlation into mechanism confirmation.

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